Cboard software. When a request is made to support a new language, translations are automatically generated from English to other languages, but they require proof-reading, especially for Indian languages. I will provide some examples to illustrate the need for proper proof-reading by individuals with language skills. Figure 2 shows a board displaying images and names of animals in English (on the left) and Malayalam (on the right). There are multiple mistakes that need to be corrected on this board alone.

For instance, the word 'chicken' is labelled as 'chicken live' (marked with a blue box on the left side of Figure 2) in the English version. Furthermore, the Malayalam board uses a transliteration of 'chicken live' in Malayalam script (marked with a blue box on the right side of Figure 2). Now, consider the word 'stable' in the English version (marked with a red box on the left side of Figure 2). The word 'stable' has two different meanings in English: one referring to something "firmly fixed" and the other to "a building set apart and adapted for keeping horses." From the image shown on the English board, it is clear that the intended meaning is the latter. However, in the Malayalam board, the incorrect translation is provided (marked with a red box on the right side of Figure 2).

Since there are numerous boards available in Cboard, a large number of volunteers are needed for proof-reading. This also demonstrates how individuals with limited technical knowledge can contribute to the development and adoption of free and open source software. Cboard uses the services of Crowdin, a proprietary, cloud-based localisation technology and services company, for the translation of boards. Crowdin facilitates the crowdsourcing of efforts from a large number of volunteers to proof-read the boards.

Cboard also uses proprietary technology for text-to-speech services, including services like Google text-to-speech, Microsoft Azure text-to-speech, and IBM Watson text-to-speech. Additionally, it leverages the built-in or add-on text-to-speech features of web browsers.

I observed some problems with the Malayalam audio rendered by Cboard. Most importantly, the pronunciation is often anglicised and difficult to understand at times. This could be an issue when training children affected by Down syndrome, autism, and cerebral palsy, who often also suffer from hearing impairments. The quality of text-to-speech in Indian languages can be improved by using larger audio datasets for training. This highlights how we can contribute to the open source community by developing audio datasets in Indian languages for training AI-based applications. Additionally, keep in mind that any issue affecting the Malayalam version of Cboard will likely affect the versions in other Indian languages as well.

Children with Down syndrome, autism, and cerebral palsy can also benefit from analysing their pupil movement while looking at a computer or smartphone screen. There are gaze tracking software available that can do this. Moreover, it can be integrated with AAC systems. Children can use their gaze to select symbols, words, or images on a screen to communicate, improving their ability to express themselves. Gaze tracking can be used to monitor and analyse eye movement patterns, providing insights into a child's focus and attention. This data can help therapists develop targeted interventions to enhance focus and reduce behavioural issues. Gaze tracking can also be used in educational games designed to engage children.

However, my search for such free and open source software was not very promising. Many such software that can be used with normal webcams are no longer maintained regularly. The ones that are actively maintained, such as Haytham Gaze Tracker, require infrared-ready webcams. However, most standard laptop webcams are not designed for infrared imaging. They are typically designed to capture visible light in well-lit environments and lack the specialised hardware needed for infrared imaging. This makes such software unusable without additional equipment.

The research for gaze tracking software also led me to the world of specialised hardware for gaze tracking, in particular, and assistive technology, in general. I won't mention them here as they are commercial and proprietary. However, it was truly eye-opening to learn about all the advanced technology (hardware and software) being used to train children with neurodevelopmental challenges in many developed countries. This technology in the Western world highlights two things: one, we in India need to catch up significantly in assistive technology to make it world class; and two, there is a pressing need for open source hardware designs that can be adopted at a far lower cost.

To sum up, free and open source software can help immensely in providing assistive technology, and people with minimal technical background can also contribute to the development of such software. END

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By: Dr Deepu Benson

The author is assistant professor at Chinmaya Vishwa Vidyapeeth (deemed to be university), Ernakulam, Kerala. He is a free software enthusiast, and is interested in theoretical computer science.